CCD: Capturing Cross-Correlations with Deformable Convolutional Networks for Multivariate Time Series Forecasting

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Abstract

The accuracy of Multivariate Time Series Forecasting relies on capturing precise channel correlations. Recent studies have primarily focused on capturing distinct channel relationships within each individual frequency and scale. However, complex channel dependencies are more prominently found in correlations across scales and frequencies. We term these relationships as *Cross-Correlations*. Drawing inspiration from Computer Vision, an efficient and effective strategy for capturing such 2D dependencies is to represent time series as *Channel-Frequency* and *Channel-Scale* views that possess image-like structures, and then use Convolutional Neural Networks to capture these complex *Cross-Correlations*.

However, it is challenging to capture *Cross-Correlations* as the correlated parts are often non-local distributed in the views. For this challenge, we propose the *Adaptive 2-Dimensional Shuffle* that permutes the rows and columns of the *Channel-Frequency* and *Channel-Scale* views before applying CNNs. Nevertheless, these views still lack well-defined spatial properties. To address this challenge, we further propose a novel *Cross Deformable Convolution*. In this module, we design a more data-adaptive kernel shape with enhanced robustness and scalability, and introduce a dynamic sparse interaction mechanism to effectively capture complex *Cross-Correlations*. In summary, we propose a novel framework called **CCD**, which **C**aptures **C**ross-Correlations with **D**eformable Convolutional Networks for Multivariate Time Series Forecasting. Extensive experiments on real-world datasets demonstrate the state-of-the-art performance of CCD.

CCS Concepts

• **Computing methodologies** → **Machine learning**.

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Keywords

Multivariate time series forecasting, Cross-correlation modeling, Deformable convolutional networks

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1 Introduction

Multivariate Time Series Forecasting (MTSF) has been studied significantly in various domains, ranging from energy management [13, 15, 24, 66, 73, 81], traffic accident prevention [23, 31, 45, 84, 85, 87, 88, 92], and extreme disaster warning [1, 35, 39, 49, 63, 70, 82]. The accuracy of MTSF relies heavily on capturing precise channel correlations, as these correlations significantly influence the quality of predictions [14, 37, 47, 54, 55, 65].

Accurately capturing correlations for multivariate time series is a challenging task due to the intrinsic multi-frequency and multi-scale nature inherent to time series [10, 28, 59, 61]. The correlations among channels exist in various ways. Specifically, as shown in Figure 1a and Figure 1b, the *Intra-Correlations* refer to the dependencies that exist among channels at the same frequency or scale, while the *Cross-Correlations* refer to the dependencies that exist among channels and across frequencies or scales. *Intra-Correlations* is a special case of *Cross-Correlations*. Recent studies on multivariate time series primarily focus on the *Intra-Correlations* that capture distinct channel relationships within each frequency [51, 83, 94] or scale [4, 8, 27]. However, they neglected *Cross-Correlations*, which are potentially more significant. In Figure 2, we plot a heat map of the dependencies among channels from a real-world dataset. The diagonal regions represent *Intra-Correlations*, while the off-diagonal regions represent *Cross-Correlations*. As can be seen, in some cases, these *Cross-Correlations* are even more significant than their *Intra-Correlations* counterparts. For example, the dependency case$_1$ between channel C_2 at scale S_1 and channel C_3 at scale S_1

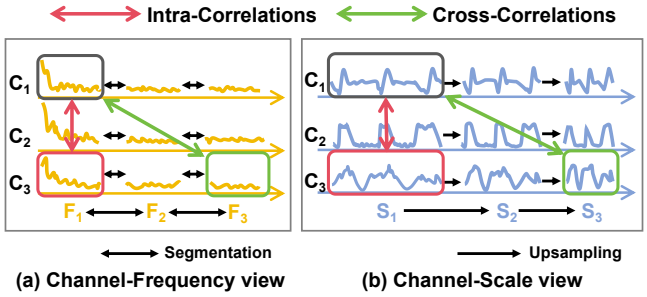


Figure 1: (a) After applying the Fourier Transform, we segment the spectrum into non-overlapping patches and reconstruct it as a 3D tensor of the form [Frequency Patch Index, Channel Index, In-Patch Data Points], creating a structure analogous to a multi-channel image. (b) After the upsampling operation, we reconstruct the time series as a 3D tensor of the form [Scale Index, Channel Index, In-Scale Data Points], creating a structure analogous to a multi-channel image.

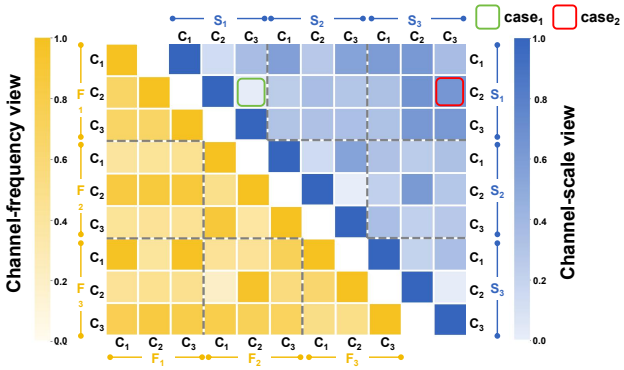


Figure 2: Heatmap of *Cross-Correlations* in *Channel-Frequency* views (bottom-left) and *Channel-Scale* views (top-right) based on a real dataset. The details of the metrics are listed in Appendix A

may be less evident than case$_2$ between channel C_2 at scale S_1 and channel C_3 at scale S_3 in Figure 2.

The intricate *Cross-Correlations* among channels collectively present a challenging 2-dimensional problem. Inspired by the remarkable success achieved in the Computer Vision for capturing such 2-dimensional dependencies [5, 20, 44, 56, 57, 62, 68], we transform time series into *Channel-Frequency* and *Channel-Scale* views with image structure (as shown in Figure 1). These views offer a richer temporal feature representation and a more intuitive layout for 2-dimensional *Cross-Correlations* than the original time series format. In this context, applying Convolutional Networks (CNNs) [25, 33, 64, 67] to the *Channel-Frequency* and *Channel-Scale* views has proven to be an efficient and effective strategy. Nevertheless, when processing image-like structures inherently embedded with temporal information, it encounters the following challenge:

It is challenging to capture *Cross-Correlations* as the correlated parts are often non-local distributed in the views. Due to the collection and structure of the time series, in *Channel-Frequency* and *Channel-Scale* views, adjacent indices do not necessarily imply correlations. Instead, two parts that do have a genuine *Cross-Correlation* are often dispersed far outside the local vicinity. Due to this non-local distribution, directly applying CNNs to *Channel-Frequency* and *Channel-Scale* views yields suboptimal accuracy, as the receptive fields may capture a mixture of uncorrelated features rather than meaningful local patterns. Although some methods based on large-kernel [7, 19, 91] and dilated convolutions [9, 77, 95] can enlarge the receptive field to encompass critical dependencies, this expansion simultaneously incorporates more noise and redundant information, which can degrade model performance.

To address this challenge, we propose the *Adaptive 2-Dimensional Shuffle* that permutes the rows and columns of the *Channel-Frequency* and *Channel-Scale* views before applying CNNs. In the resulting reordered view, correlated parts are more likely to be placed in adjacent positions, thereby mitigating the non-local distribution inherent to the original view. In this module, we use global indicators to generate a robust ranking and design a method to address the problem of gradient discontinuity caused by the discrete 2-dimensional shuffle operation. Although this data reorganization mitigates the non-local distribution of correlated parts, enabling CNNs to overcome the limitation of their local receptive fields, it still faces the following problem.

It is challenging to capture *Cross-Correlations* as these views lack well-defined spatial properties. In contrast to natural images, *Channel-Frequency* and *Channel-Scale* views lack geometric properties and involve two dimensions with distinct meanings, creating an inherent asymmetry. Although the shuffle operation arranges correlated parts in adjacent positions, these views still lack well-defined spatial properties due to the aforementioned characteristics, thereby limiting the performance of CNNs. With their dynamic modeling capabilities, methods based on Deformable Convolution [6, 17, 100] and Convolutional Attention [12, 18, 58] have the potential to solve this problem. Specifically, they adaptively adjust the convolution kernel's sampling locations or sampling weights, allowing for a more flexible capturing of dependencies. However, the lack of well-defined spatial properties complicates the learning of dynamic sampling locations [6] and causes attention mechanisms to overfit to noise or spurious correlations in the training data [38], thereby impairing generalization.

For this challenge, we further design the *Cross Deformable Convolution* to capture *Cross-Correlations*. In this module, we first introduce the *Non-uniform Cross-Extension initialization*. This initialization approach allows the model to independently and asymmetrically expand the receptive field along two dimensions, while also performing non-uniform sampling to better adapt to the shuffled views, thereby reducing the difficulty of learning sampling locations. To further enhance model robustness, we propose *Dynamic Mask-based modulation*. This modulation implements dynamic sparsification by learning to mask out noisy or irrelevant features.

In summary, we proposing a novel framework, **CCD**, which **C**aptures ***C**ross-Correlations* with **D**eformable convolutional networks for multivariate time series forecasting. Experimental results show that the proposed CCD achieves SOTA performance on real-world forecasting datasets. Our contributions are summarized as:

- We present CCD, a novel framework that simultaneously captures *Cross-Correlations* in *Channel-Frequency* and *Channel-Scale* views to enhance multivariate time series forecasting.
- We design the *Adaptive 2-Dimensional Shuffle*, which reorders the rows and columns of the *Channel-Frequency* and *Channel-Scale* views. This process creates local continuity within these views, enabling a more effective capture of dependencies.
- We propose *Cross Deformable Convolution*, which incorporates two key enhancements: Non-uniform Cross Extension initialization and Dynamic mask-based modulation. These enhancements improve the capability of the model to handle these views.
- Experimentally, CCD achieves state-of-the-art performance on multivariate time series datasets from multiple domains. Our code is available at https://github.com/decisionintelligence/CCD.

2 Related works

2.1 Multivariate Time Series Forecasting

Multivariate Time Series Forecasting (MTSF) is the task of predicting future values of multiple time-dependent variables using their shared history. The field has evolved significantly, progressing from foundational statistical models (e.g., ARIMA [3], VAR [22]) and classical machine learning algorithms (e.g., LightGBM [30], Random Forest [22]) to the current state-of-the-art, which is dominated by advanced deep learning architectures like PatchTST [48], Pathformer [10], and DUET [54]. To enhance the modeling capabilities of deep learning models for time series data, recent work has focused on designing novel networks to capture the complex dynamic characteristics and intrinsic patterns of time series from multiple dimensions, such as the frequency and multi-scale perspectives.

Frequency-Domain Modeling Frequency-domain models exploit frequency information to capture periodic patterns and global trends. As the field of temporal prediction continues to evolve, an increasing number of models are transitioning from learning time-domain signals to focusing on frequency-domain learning. For example, FreTS [94] enhances MLPs by applying frequency with energy compaction. FEDformer [99] integrates Fourier-based convolution with decomposition mechanisms for better trend-seasonal modeling. FITS [89] operates entirely in the complex frequency domain, leveraging interpolation over complex value's components rather than processing raw time-domain sequences.

Multi-scale Modeling Real-world time series often consist of various underlying patterns. To leverage the characteristics of different patterns, recent approaches tend to decompose the series into multiple scales. SCINet [41] utilizes a hierarchical downsampling tree to iteratively extract and exchange multi-scale information. Pyraformer [42]constructs a pyramidal graph of different scales and employs a pyramid attention mechanism to extract multi-resolution representations. MICN [72] processes different scales separately through multiple branches with distinct convolution kernels and subsequently merges the outputs. Inspired by hierarchical forecasting, Scaleformer [59] iteratively refine the outputs from coarser to finer scales. Additionally, TimeMixer [75] and AMD [26] thoroughly explore MLP-based interaction methods with multi-scale time-series, balancing both effectiveness and efficiency.

However, existing methods merely explore the relationships between different frequencies or scales within a single channel, failing to deeply investigate the more important relationships between different frequencies or scales across different channels.

2.2 Channel dependencies in MTSF

To overcome the challenges of modeling dependencies, researchers have developed advanced techniques to model inter-channel dependencies. These approaches can be summarized as the following categories: **1) Attention-based methods:** For instance, Crossformer [96] and iTransformer [43] utilize channels as tokens for Transformer. On this basis, DUET [54] addresses the partial dependency among channels by masking attention scores. Furthermore, Fredformer [51] transforms time series into the frequency domain and captures *Intra-Correlations* within each frequency. However, it overlooks the information-rich *Cross-Correlations*. **2) Graph neural networks:** Some studies model inter-channel relationships using various graph structures, such as simple graphs [4, 27, 60, 86, 90], hypergraphs [80], and dynamic graphs [93, 97]. Among these, MSGNet [4] utilizes multi-scale modeling, to capture the *Intra-Correlations* within each scale. However, this approach neglects the potentially more significant *Cross-Correlations*. **3) Convolutional neural networks:** Many models, such as TimesNet [78], and ModernTCN [46], employ 1D or 2D convolution along the temporal dimension and fuse information across channels, which results in poor robustness and inflexible modeling of channel interactions. **4) Others:** Another line of work employs MLP-based channel mixing, with notable examples being TSMixer [11], TimeMixer [75], and Tiny Time Mixers [21]. Furthermore, following the recent success of Mamba-based methods in the time series domain, TimePro performs variable interactions directly on the Hyper-state of the Mamba architecture.

The aforementioned methods merely consider channel relationships within one single frequency or scale, overlooking the more prevalent *Cross-Correlations* that occur across different frequencies or scales. To close this gap, we propose a novel framework to effectively capture these complex *Cross-Correlations*.

3 Methodology

3.1 Problem Statement

Given a historical multivariate time series input $\boldsymbol{X} \in \mathbb{R}^{N \times T}$ the aim of MTSF is to predict the future series $\boldsymbol{Y} \in \mathbb{R}^{N \times F}$, where N denotes the number of channels, T and F represent the lookback and forecast horizons, respectively.

3.2 Structure Overview

Figure 3 shows the architecture of CCD, which structures the time series into *Channel-Scale* (CS) and *Channel-Frequency* (CF) views and utilizes a deformable convolution-based method to capture complex *Cross-Correlations*. Specifically, we employ *CS View Generator* and *CF View Generator* to transform the input series into image structures. Modeling on these specially designed views can capture complex *Cross-Correlations* on both *CF* and *CS* views. Subsequently, the same operations are applied to both views in parallel.

First, we introduce the Adaptive 2-Dimensional Shuffle. This module utilizes *global indicators* from both the frequency (or scale) and channel dimensions to adaptively generate a robust ordering. Furthermore, we propose an effective method to ensure proper

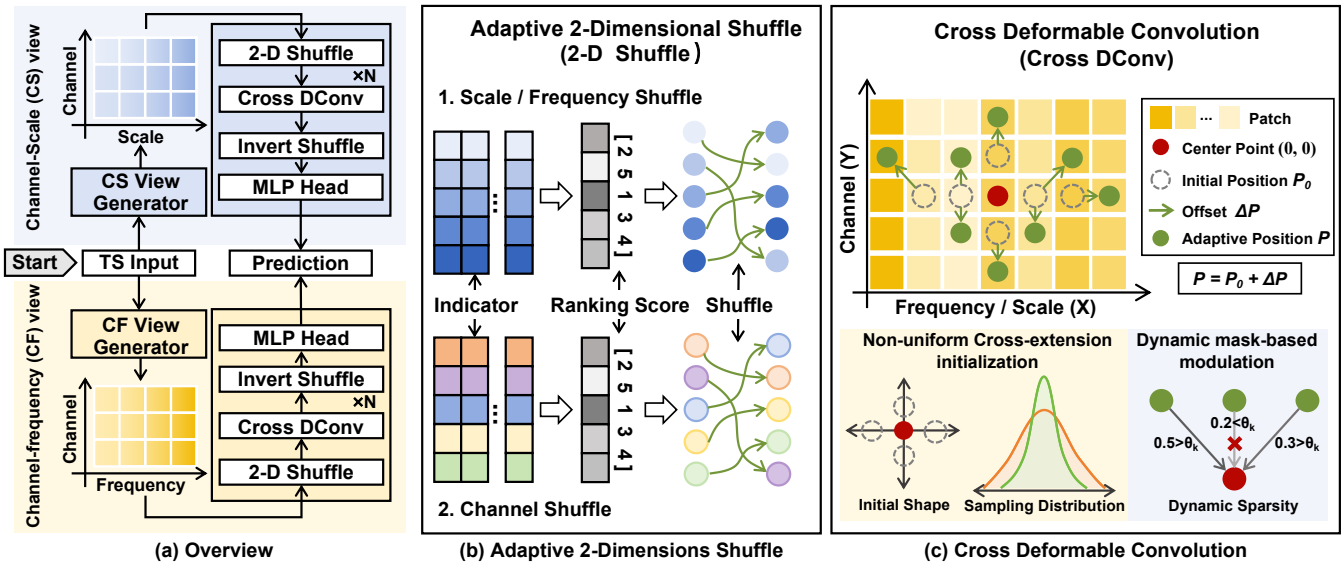


Figure 3: The overall framework of CCD. The CS and CF View Generators transform time series into image-like structures and embed them into the feature space. *Adaptive 2-Dimensional Shuffle* rearranges the data across two dimensions, mitigating the non-local distribution of correlated parts. *Cross Deformable Convolution* applies a 2D deformable convolution which enhanced by *Non-uniform Cross-extension initialization* and *Dynamic mask-based modulation*, to the shuffled views to capture complex dependencies. Finally, we restore the original sequence, which is then fed into *MLP Heads* to generate the final prediction.

gradient propagation during the 2-dimensional shuffle. In this way, we reorder the discrete *CF* and *CS* views, adaptively arranging patches with potential dependencies in adjacent positions.

Afterwards, *Cross Deformable Convolution* is performed on the shuffled views to extract the frequency (or scale) information and capture the complex *Cross-Correlations*. We design two key enhancements to the standard deformable convolution. The first is *Non-uniform Cross-extension initialization*. This approach allows the model to independently and asymmetrically expand the receptive field along two dimensions, while also performing non–uniform sampling to better adapt to the shuffled views. The second enhancement is Dynamic mask-based modulation, a technique that learns to generate a dynamic mask selectively nullifying noisy and irrelevant features. Benefit from these enhancements, the *Cross Deformable Convolution* is better at capturing complex *Cross-Correlations*

Finally, we restore the original order and generate predictions based on MLPs in the two branches. Below, we will introduce the details of each module in our framework.

3.3 View Generator

Scale analysis and frequency analysis reveal the structural characteristics and behavior patterns of data at different levels, which is crucial for a deep understanding of time series [28, 74, 75]. Thus, we further explore their deep coupling with channel dependencies to capture *Cross-Correlations* and enhance the predictions. To this end, we first use *CF View Generator* and *CS View Generator* to preliminarily extract time series features and reorganize them into *Channel-Scale* and *Channel-Frequency* views.

3.3.1 CS View Generator. To avoid using average pooling, which can lead to information loss [74], we employ a series of parallel 1D convolutional operations with kernel size as $\{d_0, d_1, \ldots, d_{M_s}\}$ to extract multi-scales series $\{s_{i,j} \in \mathbb{R}^{\lfloor \frac{T}{d_j} \rfloor} \mid i \in [1, N], j \in [1, M_s]\}$, where $s_{i,j}$ denotes the series of i-th channel at j-th scale. Subsequently, we employ M_s linear layers to embed them at the series level, obtaining the *Channel-Scale* view $\boldsymbol{CS} \in \mathbb{R}^{N \times M_s \times dims}$ with a hidden dimension of *dims*.

3.3.2 CF View Generator. To decouple different frequencies, we first divide the data into M_f non-overlapping patches after applying a Fast Fourier Transform (FFT). The patched data can be expressed as $F \in \mathbb{C}^{N \times M_f}$. We then use a Linear layer to embed it at the patch level and get the *Channel-Frequency* view $\boldsymbol{CF} \in \mathbb{R}^{N \times M_f \times dims}$, where *dims* is the size of the hidden dimension.

3.4 Adaptive 2-Dimensions Shuffle

To address the issue that the correlated parts are often non-locally distributed in *CF* and *CS* views, we propose the *Adaptive 2-Dimensions Shuffle* module (Figure 3b). Guided by *global indicators*, this module adaptively generates robust *ranking scores* that determine the order for both the scale (or frequency) and channel dimensions. After the *shuffle* operation, adjacent patches are more likely to have underlying dependencies, thereby enabling convolutional networks to directly capture 2-Dimensional dependencies. Since identical operations are subsequently performed on both the views $\boldsymbol{CS}$ and $\boldsymbol{CF}$, we use the $\boldsymbol{CF}$ view as an illustrative example.

To mitigate the impact of short-term noise on rank learning of channels and frequencies, we propose a novel approach incorporating *global indicators* $\boldsymbol{I}_c \in \mathbb{R}^{N \times dims}, \boldsymbol{I}_f \in \mathbb{R}^{M_f \times dims}$. Leveraging

a long-term perspective, these indicators enable the learning of a robust and generalizable ranking.

By summing the indicator dimensions into $\boldsymbol{score}_c \in \mathbb{R}^N$ and $\boldsymbol{score}_f \in \mathbb{R}^M_f$, we obtain the *ranking scores* of the corresponding channels and frequencies. We then employ argmax to calculate the target permutation order $\boldsymbol{\tau}_c$ and $\boldsymbol{\tau}_f$ for shuffle:

$$\boldsymbol{\tau}_c = \text{argsort}(\boldsymbol{score}_c) \in \mathbb{R}^N, \tag{1}$$

$$\boldsymbol{\tau}_f = \text{argsort}(\boldsymbol{score}_f) \in \mathbb{R}^{M_f}. \tag{2}$$

We aim to perform sorting based on $\boldsymbol{\tau}_c$ and $\boldsymbol{\tau}_f$, but the discreteness and non-differentiability of standard sorting operations preclude their direct use in gradient-based optimization. As a solution, soft ranking methods [2, 69] offer a differentiable approximation by computing a probability distribution for the relative order of elements. However, this approach can introduce noise and sacrifice the precision and intuitive clarity of the sorting outcome.

To address this, we propose a method that maintains gradient propagation and remains accurate, similar to traditional methods. We sequentially apply $\boldsymbol{\tau}_c$ and $\boldsymbol{\tau}_f$ to permute $\boldsymbol{CF}$, obtaining the view $\boldsymbol{CF}'$ with the learnable orders. We also apply the same permutations to $\boldsymbol{scores}_c$ and $\boldsymbol{scores}_f$ to derive $\boldsymbol{scores}'_c$ and $\boldsymbol{scores}'_f$ for subsequent correct gradient propagation. As $\boldsymbol{CF}'$ loses the gradient due to the non-differentiable permutation operation, we explicitly define the gradient flow during the backward pass by injecting the gradients of $\boldsymbol{scores}_c$ and $\boldsymbol{scores}_f$, effectively employing a straight-through proxy to ensure adaptive rank learning:

$$\boldsymbol{grad}_c = \boldsymbol{score}'_c \odot \text{detach}(\frac{1}{\boldsymbol{score}'_c}) \in \mathbb{R}^{N\times dims}, \tag{3}$$

$$\boldsymbol{grad}_f = \boldsymbol{score}'_f \odot \text{detach}(\frac{1}{\boldsymbol{score}'_f}) \in \mathbb{R}^{M_f\times dims}, \tag{4}$$

$$\boldsymbol{CF}_{shuf} = \boldsymbol{CF}' \odot \boldsymbol{grad}_c \odot \boldsymbol{grad}_f \in \mathbb{R}^{N\times M_f\times dims}, \tag{5}$$

where $\boldsymbol{grad}_c$ and $\boldsymbol{grad}_f$ are all-one matrix with gradients, detach() is a method to cut off the gradient, and $\boldsymbol{CF}_{shuf}$ is the final shuffled view. We record $\boldsymbol{\tau}_c$ and $\boldsymbol{\tau}_f$ during the shuffle process to enable the inverse shuffle.

3.5 Cross Deformable Convolution

Deformable Convolution [17] dynamically adapts kernel shapes to capture geometric transformations effectively. Specifically, for each initial position $\boldsymbol{P_0}$ on the regular sampling grid of a convolution window, the module predicts a data-dependent offset $\boldsymbol{\Delta P}$ from the input features. By adding this offset, the sampling point is shifted to a new adaptive position $\boldsymbol{P} = \boldsymbol{P_0} + \boldsymbol{\Delta P}$.

To better capture *Cross-Correlations* in views which lack well-defined spatial properties, we propose the *Cross Deformable Convolution* (Figure 3c). It enhances standard deformable convolution by introducing *Non-uniform Cross-extension initialization* to accommodate specialized image-like views and *Dynamic mask-based modulation* for robust interaction, thereby optimizing its effectiveness on the *Channel-Frequency* and *Channel-Scale* views.

For descriptive convenience, we establish a 2D grid coordinate system on the views and assume a given convolutional window is centered at $(0, 0)$, which is employed to capture patches correlated with the center patch. In this system, the x-axis and y-axis are aligned with the frequency and channel dimensions, respectively.

3.5.1 Sampling with initialization. The standard initialization strategy of Deformable Convolution utilises a square shape and uniform sampling. With the receptive field size as r, the initial shape S_{square} is defined as:

$$\boldsymbol{S}_{square} = \{(x, y) \in \mathbb{Z}^2 \mid |x|, |y| \le r\}. \tag{6}$$

This square-shaped initialization leads to a quadratic scaling of sampling points relative to the receptive field, incurring significant computational overhead and noise. To efficiently expand the receptive field for capturing complex *Cross-Correlations*, we devise a method that employs a cross-shaped initialization, which extends linearly along the frequency (x) and channel (y) dimensions, respectively. The layout is formally defined as follows:

$$\boldsymbol{S}_{cross} = \{(x, 0) \in \mathbb{Z}^2 \mid |x| \le r_f\} \cup \{(0, y) \in \mathbb{Z}^2 \mid |y| \le r_c\}, \tag{7}$$

where r_f, r_c are the receptive field size in frequency and channel dimensions, respectively. This approach enables the model to independently and asymmetrically expand its receptive field along two dimensions, which only linearly increases the number of sampling points. More importantly, it allows the model to capture *Intra-Correlations* along frequency (x) and channel (y) by default, and building upon this, explore cross dependencies.

Furthermore, the standard strategy initializes one sampling point for each position $\boldsymbol{s} \in \boldsymbol{S}_{square}$, which widens the gap with the uneven distribution of *Cross-Correlations*. To address this, we design a non-uniform sampling strategy that allocates a varying number of sampling points across different position $\boldsymbol{s} \in \boldsymbol{S}_{cross}$. Specifically, the number of samples num_s at position $\boldsymbol{s}$ is determined by the distance from the center point $\|\boldsymbol{s}\|$, following a normal distribution:

$$num_s = \max(\lfloor \frac{num_{max}}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(\|\boldsymbol{s}\| - \mu)^2}{2\sigma^2}\right) \rfloor, 1), \tag{8}$$

where num_{max},σ and μ are hyperparameters that adjust the normal distribution. The collection of initial positions for all sampling points across all convolutional windows is formally defined as $\boldsymbol{P}_0 \in \mathbb{R}^{N\times M_f\times(num\times 2)}$, where $num = \sum_{s\in S_{cross}} num_s$ denote the number of total sampling points for each center point. This strategy encourages the *Adaptive 2D Shuffle* to relocate more correlated patches to positions nearer to the grid center.

Following initialization, a 2D Depthwise Separable Convolution ϕ_1 [16, 29] is applied to generate the offsets $\boldsymbol{\Delta P}$, which are subsequently used to compute the adaptive sampling position $\boldsymbol{P}$:

$$\boldsymbol{\Delta P} = \phi_1(\boldsymbol{CF}_{shuf}) \in \mathbb{R}^{N\times M_f\times(num\times 2)}, \tag{9}$$

$$\boldsymbol{P} = (\boldsymbol{P}_0 + \boldsymbol{\Delta P}) \in \mathbb{R}^{N\times M_f\times(num\times 2)}. \tag{10}$$

Since the adaptive positions $\boldsymbol{P}$ are typically fractional and do not align with the discrete grid points of the feature map, we use Bilinear Interpolation [32], denoted as ψ, to extract the features $\boldsymbol{Z}$ at these precise positions for interaction with the center point:

$$\boldsymbol{Z} = \psi(\boldsymbol{CF}_{shuf}, \boldsymbol{P}) \in \mathbb{R}^{N\times M_f\times num\times dims}. \tag{11}$$

3.5.2 Adaptive interaction. Given that the number of effective dependencies varies across different convolutional windows, we propose *Dynamic mask-based modulation* for the *Cross Deformable*

Convolution. This mechanism adaptively masks irrelevant interactions between center and samplings points in modulation coefficients $\boldsymbol{W}$ based on the adaptive threshold $\boldsymbol{\tau}$, enabling the model to dynamically capture sparse dependencies.

We employ additional convolutions to learn the modulation coefficients $\boldsymbol{W}$ and the mask threshold $\boldsymbol{\Gamma}$:

$$\boldsymbol{W} = \phi_2(\boldsymbol{CF}_{shuf}) \in \mathbb{R}^{N\times M_f\times num}, \tag{12}$$

$$\boldsymbol{\Gamma} = \phi_3(\boldsymbol{CF}_{shuf}) \in \mathbb{R}^{N\times M_f\times 1}, \tag{13}$$

where ϕ_2 and ϕ_3 are the same convolutions like ϕ_1. To ensure gradient propagation, we employ a differentiable soft mask, which suppresses the values in $\boldsymbol{W}$ that are below the mask threshold towards zero, yielding the masked modulation coefficients $\boldsymbol{W}_{masked}$:

$$\boldsymbol{W}_{inf} = \log(\text{sigmoid}((\boldsymbol{W} - \boldsymbol{\Gamma})/0.01)), \tag{14}$$

$$\boldsymbol{W}_{masked} = \text{softmax}(\boldsymbol{W}_{inf} + \boldsymbol{W}). \tag{15}$$

Finally, we perform feature interaction between the sampling points and their corresponding center points to capture *Cross-Correlations*, modulated by the coefficients $\boldsymbol{W}_{masked}$:

$$\boldsymbol{Z}_{modulated} = \boldsymbol{Z} \odot \boldsymbol{W}_{masked} \in \mathbb{R}^{N\times M_f\times num\times dims}, \tag{16}$$

$$\boldsymbol{Z}_{combined} = \text{sum}(\boldsymbol{Z}_{modulated}) \in \mathbb{R}^{N\times M_f\times dims}, \tag{17}$$

$$\boldsymbol{CF}_{out} = (\boldsymbol{Z}_{combined} + \boldsymbol{CF}_{shuf}) \in \mathbb{R}^{N\times M_f\times dims}. \tag{18}$$

3.6 Prediction

After applying the Cross Deformable Convolution, we separately apply *Invert Shuffle* operation to $\boldsymbol{CS}_{out}$ and $\boldsymbol{CF}_{out}$ to restore their original order to get $\boldsymbol{CS}_{invert}$, $\boldsymbol{CF}_{invert}$, and then make predictions on each stream independently.

3.6.1 Channel-Scale Prediction. Unlike most existing methods [10, 48, 54], we employ a more expressive MLP-based prediction head in place of a linear one to enhance the predictive capabilities:

$$\hat{\boldsymbol{Y}}_s = \text{MLP}_s(\boldsymbol{CS}_{invert}) \in \mathbb{R}^{N\times F}, \tag{19}$$

where $\text{MLP}_s = \mathbb{R}^{(Ms\times dims)} \to \mathbb{R}^F$ is used to transform the flattened $\boldsymbol{CS}_{invert}$ to prediction.

3.6.2 Channel-Frequency Prediction. We first predict the data in the frequency domain, and then transform the result back to the time domain and refine the prediction:

$$\boldsymbol{CF}_{real} = \text{MLP}_{real}(\boldsymbol{CF}_{invert}) \in \mathbb{R}^{N\times F}, \tag{20}$$

$$\boldsymbol{CF}_{imag} = \text{MLP}_{imag}(\boldsymbol{CF}_{invert}) \in \mathbb{R}^{N\times F}, \tag{21}$$

where $\text{MLP}_{real}, \text{MLP}_{imag} = \mathbb{R}^{(M_f\times dims)\to\mathbb{R}^F}$. We then combine $\boldsymbol{CF}_{real}$ and $\boldsymbol{CF}_{imag}$ and into complex data and perform iFFT operations into $\hat{\boldsymbol{Y}}_{real}$ and $\hat{\boldsymbol{Y}}_{imag}$:

$$\hat{\boldsymbol{Y}}_f = \text{Linear}_f(\text{concat}(\hat{\boldsymbol{Y}}_{real}, \hat{\boldsymbol{Y}}_{imag})) \in \mathbb{R}^{N\times F}, \tag{22}$$

where $\text{Linear}_f = \mathbb{R}^{2\times F} \to \mathbb{R}^F$ is used to refine the prediction in time-domain. The final prediction is:

$$\hat{\boldsymbol{Y}} = \hat{\boldsymbol{Y}}_s + \hat{\boldsymbol{Y}}_f. \tag{23}$$

4 Experiments

4.1 Experimental Settings

4.1.1 Dataset. To conduct comprehensive and fair comparisons for different models, we conduct experiments on eight well-known forecasting benchmarks as the target datasets, including: (I) **ETT** [98] datasets contain 7 variates collected from two different electric transformers from July 2012 to July 2014. It consists of four subsets, of which ETTh1/ETTh2 are recorded hourly, and ETTm1/ETTm2 are recorded every 15 minutes. (II) **Electricity** [71] contains the electricity consumption of 321 customers from 2012 to 2014, recorded hourly. (III) **Traffic** [79] contains road occupancy rates measured by 862 sensors on freeways in the San Francisco Bay Area from 2015 to 2016, recorded hourly. (IV) **Solar** [34] records solar power generation from 137 PV plants in 2006, every 10 minutes. (V) **Weather** [79] collects 21 meteorological indicators, including temperature and barometric pressure, for Germany in 2020, recorded every 10 minutes. More details of the benchmark datasets are included in Table 1.

4.1.2 Baselines. We choose the state-of-the-art models in recent years, to serve as baselines, including DUET [54], TimePro [47], TimeKAN [28], AMD [26], Fredformer [51], MSGNet [4], iTransformer [43], PatchTST [48], Crossformer [96], and TimesNet [78]. These methods encompass architectures based on CNN, Transformer, GNN, MLP, KAN, and Mamba, which are designed to model distinct data dimensions: frequency is modeled by TimeKAN and Fredformer; scale by AMD and MSGNet; and the channel by models such as DUET, TimePro, and iTransformer.

4.1.3 Implementation Details. To keep consistent with previous works, we adopt Mean Squared Error (MSE) and Mean Absolute Error (MAE) as evaluation metrics. We consider four forecasting horizon $F \in \{96, 192, 336, 720\}$ for all datasets and set the look-back window to 96. We utilize the TFB code repository for unified evaluation. Following the settings in TFB [53] and TSFM-Bench [36], we do not apply the 'drop-last' trick to ensure a fair comparison. We re-evaluate all the baselines under the same environment and settings for a fair comparison. All experiments of CCD are conducted using PyTorch [50] in Python 3.8 and executed on an NVIDIA Tesla-A800 GPU. Our code is available at https://github.com/decisionintelligence/CCD.

4.2 Main Results

We report the results of multivariate time series forecasting in Table 2. The input length T is set to 96 for our method and all baselines. The forecasting horizon $F \in \{96, 192, 336, 720\}$. We have the following observations: 1) Compared with forecasters of different structures, including Transformer-, CNN-, KAN-, GNN-, Mamba-, and MLP-based models, CCD achieves excellent predictive performance. Specifically, CCD obtains the best results in 25 out of 32 MSE cases and 31 out of 32 MAE cases across different forecasting horizons, demonstrating its strong and consistent forecasting ability. 2) In contrast to existing models that exploit scale or frequency information, CCD utilizes *Cross Deformable Convolution* to enable flexible, dynamic, and sparse modeling over scale and frequency dimensions. Specifically, compared with the scale modeling method AMD and the frequency modeling method TimeKAN, CCD reduces MSE/MAE by 6.73%/8.08% and 5.68%/8.68%, respectively. This demonstrates

Table 1: Statistics of multivariate datasets.

Dataset	Domain	Frequency	Lengths	Dim	Split	Description
ETTh1	Electricity	1 hour	14,400	7	6:2:2	Power transformer 1, comprising seven indicators such as oil temperature and useful load
ETTh2	Electricity	1 hour	14,400	7	6:2:2	Power transformer 2, comprising seven indicators such as oil temperature and useful load
ETTm1	Electricity	15 mins	57,600	7	6:2:2	Power transformer 1, comprising seven indicators such as oil temperature and useful load
ETTm2	Electricity	15 mins	57,600	7	6:2:2	Power transformer 2, comprising seven indicators such as oil temperature and useful load
Electricity	Electricity	1 hour	26,304	321	7:1:2	Electricity records the electricity consumption in kWh every 1 hour from 2012 to 2014
Solar	Energy	10 mins	52,560	137	6:2:2	Solar production records collected from 137 PV plants in Alabama
Weather	Environment	10 mins	52,696	21	7:1:2	Recorded every 10 minutes for the whole year 2020, which contains 21 meteorological indicators
Traffic	Traffic	1 hour	17,544	862	7:1:2	Road occupancy rates measured by 862 sensors on San Francisco Bay area freeways

Table 2: Multivariate forecasting results with an input length of $T = 96$. Results are reported under four forecasting horizons, i.e., $F \in \{96, 192, 336, 720\}$, and Avg denotes the average over these horizons. The best and second-best results are highlighted in bold and underlined, respectively.

Models		CCD (ours)		DUET (2025)		TimeKAN (2025)		AMD (2025)		TimePro (2025)		Fredformer (2024)		MSGNet (2024)		iTransformer (2024)		PatchTST (2023)		Crossformer (2023)		TimesNet (2023)	
Metrics		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	96	**0.370**	**0.387**	0.377	0.393	0.374	0.391	0.375	0.392	0.375	0.398	0.378	0.395	0.390	0.411	0.386	0.405	0.414	0.419	0.423	0.448	0.384	0.402
	192	0.424	**0.418**	0.429	0.425	**0.421**	0.421	0.430	0.422	0.427	0.429	0.435	0.424	0.443	0.442	0.441	0.436	0.460	0.445	0.471	0.474	0.436	0.429
	336	**0.464**	**0.438**	0.471	0.446	0.464	0.440	0.471	0.443	0.472	0.450	0.485	0.447	0.482	0.469	0.487	0.458	0.501	0.466	0.570	0.546	0.491	0.469
	720	0.468	**0.456**	0.496	0.480	**0.466**	0.462	0.478	0.464	0.476	0.474	0.496	0.472	0.496	0.488	0.503	0.491	0.500	0.488	0.653	0.621	0.521	0.500
	Avg	0.432	**0.425**	0.443	0.436	**0.431**	0.428	0.439	0.430	0.438	0.438	0.449	0.434	0.453	0.453	0.454	0.448	0.469	0.455	0.529	0.522	0.458	0.450
ETTh2	96	**0.280**	**0.328**	0.296	0.345	0.293	0.343	0.287	0.338	0.293	0.345	0.291	0.342	0.329	0.371	0.297	0.349	0.302	0.348	0.745	0.584	0.340	0.374
	192	**0.354**	**0.378**	0.368	0.389	0.375	0.396	0.367	0.388	0.367	0.394	0.372	0.390	0.402	0.414	0.380	0.400	0.388	0.400	0.877	0.656	0.402	0.414
	336	**0.398**	**0.415**	0.411	0.422	0.429	0.441	0.410	0.424	0.419	0.431	0.419	0.431	0.440	0.445	0.428	0.432	0.426	0.433	1.043	0.731	0.452	0.452
	720	0.416	0.435	**0.412**	**0.434**	0.466	0.468	0.421	0.440	0.427	0.445	0.431	0.450	0.480	0.477	0.427	0.445	0.431	0.446	1.104	0.763	0.462	0.468
	Avg	**0.362**	**0.389**	0.372	0.398	0.391	0.412	0.371	0.398	0.377	0.404	0.378	0.403	0.413	0.427	0.383	0.406	0.387	0.407	0.942	0.683	0.414	0.427
ETTm1	96	**0.308**	**0.339**	0.324	0.354	0.327	0.365	0.327	0.361	0.326	0.364	0.326	0.361	0.319	0.366	0.334	0.368	0.329	0.367	0.404	0.426	0.338	0.375
	192	**0.360**	**0.370**	0.369	0.379	0.363	0.387	0.366	0.383	0.367	0.383	0.363	0.384	0.377	0.397	0.377	0.391	0.367	0.385	0.450	0.451	0.374	0.387
	336	0.392	**0.392**	0.404	0.402	**0.389**	0.407	0.398	0.404	0.402	0.409	0.395	0.406	0.417	0.422	0.426	0.420	0.399	0.410	0.532	0.515	0.410	0.411
	720	0.455	**0.428**	0.463	0.437	0.457	0.445	0.464	0.437	0.469	0.446	0.456	0.441	0.487	0.463	0.491	0.459	**0.454**	0.439	0.666	0.589	0.478	0.450
	Avg	**0.379**	**0.382**	0.390	0.393	0.384	0.401	0.389	0.396	0.391	0.401	0.385	0.398	0.400	0.412	0.407	0.409	0.387	0.400	0.513	0.495	0.400	0.406
ETTm2	96	**0.169**	**0.247**	0.174	0.255	0.178	0.262	0.176	0.259	0.178	0.260	0.177	0.258	0.182	0.266	0.180	0.264	0.175	0.259	0.287	0.366	0.187	0.267
	192	**0.233**	**0.291**	0.243	0.302	0.244	0.308	0.242	0.302	0.242	0.303	0.243	0.301	0.248	0.306	0.250	0.309	0.241	0.302	0.414	0.492	0.249	0.309
	336	**0.290**	**0.328**	0.304	0.341	0.305	0.346	0.298	0.337	0.303	0.342	0.302	0.340	0.312	0.346	0.311	0.348	0.305	0.343	0.597	0.542	0.321	0.351
	720	**0.389**	**0.388**	0.399	0.397	0.402	0.404	0.396	0.394	0.400	0.399	0.404	0.398	0.414	0.404	0.412	0.407	0.402	0.400	1.730	1.042	0.408	0.403
	Avg	**0.270**	**0.313**	0.280	0.324	0.282	0.330	0.278	0.323	0.281	0.326	0.281	0.324	0.289	0.331	0.288	0.332	0.281	0.326	0.757	0.611	0.291	0.333
Weather	96	**0.154**	**0.192**	0.163	0.202	0.164	0.210	0.174	0.221	0.166	0.207	0.163	0.207	0.163	0.212	0.174	0.214	0.177	0.218	0.158	0.230	0.172	0.220
	192	**0.205**	**0.240**	0.218	0.252	0.209	0.250	0.219	0.259	0.216	0.254	0.224	0.258	0.211	0.254	0.221	0.254	0.225	0.259	0.206	0.277	0.219	0.261
	336	**0.262**	**0.283**	0.274	0.294	0.264	0.290	0.273	0.296	0.273	0.296	0.278	0.298	0.273	0.299	0.278	0.296	0.278	0.297	0.272	0.335	0.280	0.306
	720	0.344	**0.337**	0.349	0.343	**0.343**	0.342	0.349	0.345	0.351	0.346	0.357	0.350	0.351	0.348	0.358	0.349	0.354	0.348	0.398	0.418	0.365	0.359
	Avg	**0.241**	**0.263**	0.251	0.273	0.245	0.273	0.254	0.280	0.252	0.276	0.256	0.278	0.249	0.278	0.258	0.278	0.259	0.281	0.259	0.315	0.259	0.286
Solar	96	**0.186**	**0.197**	0.200	0.207	0.228	0.254	0.209	0.241	0.196	0.237	0.189	0.236	0.210	0.246	0.203	0.237	0.234	0.286	0.310	0.331	0.250	0.292
	192	**0.217**	**0.219**	0.228	0.233	0.241	0.284	0.235	0.260	0.231	0.263	0.227	0.259	0.265	0.290	0.233	0.261	0.267	0.310	0.734	0.725	0.296	0.318
	336	**0.229**	**0.236**	0.262	0.244	0.273	0.292	0.294	0.316	0.250	0.281	0.243	0.286	0.294	0.318	0.248	0.273	0.290	0.315	0.750	0.735	0.319	0.330
	720	**0.239**	**0.237**	0.258	0.249	0.279	0.297	0.275	0.308	0.253	0.285	0.250	0.285	0.285	0.315	0.249	0.275	0.289	0.317	0.769	0.765	0.338	0.337
	Avg	**0.218**	**0.222**	0.237	0.233	0.255	0.282	0.253	0.281	0.233	0.267	0.227	0.267	0.264	0.292	0.233	0.262	0.270	0.307	0.641	0.639	0.301	0.319
Electricity	96	**0.134**	**0.223**	0.145	0.233	0.174	0.266	0.147	0.251	0.139	0.234	0.148	0.242	0.165	0.274	0.148	0.240	0.195	0.285	0.219	0.314	0.168	0.272
	192	**0.152**	**0.238**	0.163	0.248	0.182	0.273	0.176	0.262	0.156	0.249	0.165	0.257	0.185	0.292	0.162	0.253	0.199	0.289	0.231	0.322	0.184	0.289
	336	**0.168**	**0.257**	0.175	0.262	0.197	0.286	0.193	0.281	0.172	0.267	0.180	0.274	0.197	0.304	0.178	0.269	0.215	0.305	0.246	0.337	0.198	0.300
	720	**0.202**	**0.284**	0.204	0.291	0.236	0.320	0.232	0.329	0.209	0.299	0.218	0.305	0.231	0.332	0.225	0.317	0.256	0.337	0.280	0.363	0.220	0.320
	Avg	**0.164**	**0.251**	0.172	0.259	0.197	0.286	0.187	0.281	0.169	0.262	0.178	0.270	0.195	0.300	0.178	0.270	0.216	0.304	0.244	0.334	0.193	0.295
Traffic	96	**0.388**	**0.233**	0.407	0.252	0.423	0.286	0.443	0.298	0.426	0.292	0.403	0.274	0.608	0.349	0.395	0.268	0.544	0.359	0.522	0.290	0.593	0.321
	192	**0.412**	**0.241**	0.431	0.262	0.442	0.295	0.496	0.323	0.439	0.298	0.429	0.289	0.634	0.371	0.417	0.276	0.540	0.354	0.530	0.293	0.617	0.336
	336	**0.430**	**0.249**	0.456	0.269	0.473	0.335	0.520	0.330	0.449	0.307	0.441	0.295	0.669	0.388	0.443	0.283	0.551	0.358	0.558	0.305	0.629	0.336
	720	0.469	**0.271**	0.509	0.292	0.481	0.357	0.540	0.344	0.475	0.309	**0.463**	0.300	0.729	0.420	0.467	0.302	0.586	0.375	0.589	0.328	0.640	0.350
	Avg	**0.425**	**0.248**	0.451	0.269	0.455	0.318	0.500	0.324	0.447	0.301	0.434	0.289	0.660	0.382	0.430	0.282	0.555	0.361	0.550	0.304	0.620	0.336

the effectiveness of our flexible modeling strategy. 3) CCD also demonstrates the importance of *Cross-Correlations*. Compared with state-of-the-art methods that capture *Intra-Correlations* within each scale or frequency, i.e., MSGNet and Fredformer, CCD achieves MSE reductions of 14.77%/3.76% and MAE reductions of 13.25%/6.38%, respectively. Furthermore, compared with the state-of-the-art channel correlation modeling methods DUET and TimePro, CCD reduces MSE by 4.05%/3.69% and MAE by 3.47%/6.74%, respectively. These results indicate that explicitly considering *Cross-Correlations* is more effective for modeling complex channel dependencies.

4.3 Model Analyses

4.3.1 Ablation Studies. To ascertain the impact of different modules within CCD, we perform ablation studies focusing on the following components:

1) w/o Channel-Scale: Remove the branch which capture the *Cross-Correlations* in *Channel-Scale* views.

2) w/o Channel-Frequency: Remove the branch *Cross-Frequency* in *Channel-Frequency* views.

3) w/o 2-D Shuffle: Remove the *Adaptive 2-Dimensional Shuffle* module in both branches.

4) Standard initialization: Replace *Non-uniform Cross-Extension initialization* with standard initialization.

5) w/o Modulation: Remove *Dynamic mask-based modulation.*

Table 3 illustrates the unique impact of each module. We have the following observations: 1) Ablation of either the *Channel-Scale* or *Channel-Frequency* view branch leads to a significant decline in performance. This finding is particularly pronounced on the ETTh2 dataset, which is smaller in scale and therefore requires a rich feature representation. This indicates the effectiveness of transforming time series into views with image structures for subsequent dependency capture. 2) When the *Adaptive 2-Dimensional Shuffle* module is removed, the *Channel-Scale* and *Channel-Frequency* views suffer from the problem of non-local distribution. This makes it difficult for the model to identify effective information, leading to a decline in performance. This highlights the necessity of *Adaptive 2-Dimensional Shuffle* in the *Channel-Scale* and *Channel-Frequency* views. 3) When our proposed *Non-uniform Cross-Extension initialization* is replaced with standard initialization (which initializes sampling points in a square shape and ensures uniform sampling), learning the sampling point offsets becomes difficult. Consequently, the model's performance degrades across all datasets, which demonstrates the effectiveness of our initialization strategy. This point is further corroborated in Section 4.3.4. 4) When the *Dynamic mask-based modulation* is removed, the model no longer selects samples dynamically. Instead, it statically averages all sampling points within the convolutional window. This approach invariably incorporates more noise, which impairs overall performance. This suggests that adaptive dynamic interaction is highly effective for enhancing prediction accuracy.

Table 3: Ablation Study of CCD. The input length T is 96. All results are averaged across four different forecasting horizons: $F \in \{96, 192, 336, 720\}$. The best and second-best results are highlighted in bold and underlined, respectively.

Datasets	ETTh2		ETTm2		Weather		Solar	
Metrics	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
w/o Channel-Scale	0.386	0.399	0.275	0.315	0.246	0.266	0.228	0.230
w/o Channel-Frequency	0.377	0.396	0.281	0.320	0.250	0.267	0.230	0.235
w/o 2-D Shuffle	0.392	0.413	0.289	0.332	0.256	0.279	0.235	0.238
w/o Initialization	0.383	0.401	0.276	0.316	0.248	0.268	0.227	0.228
w/o Modulation	0.378	0.397	0.276	0.316	0.245	0.267	0.228	0.227
CCD	**0.362**	**0.389**	**0.270**	**0.313**	**0.241**	**0.263**	**0.218**	**0.222**

4.3.2 Parameter Sensitivity. We study the parameter sensitivity of the CCD, including the patch numbers M_f in *Channel-Frequency* views, the hidden dimension size *dims*, the channel radius r_c and frequency radius r_f in *Cross Extension Initialization*.

Figure 4a shows the impact of the number of patches on performance. As the number of patches increases, the *Channel-Frequency* view exhibits richer *Cross-Correlations*, but the amount of noise also increases. We typically select 8 or 12 as the preferred options. Besides, Figure 4b illustrates the impact of hidden dimension size on model performance. It demonstrates that our method is not dependent on a large number of hidden dimensions to achieve strong results. To balance computational efficiency with performance, we generally opt for a dimension of 48 or 64. Figure 4c and Figure 4d present that our model is robust to an increasing receptive field radius. We attribute this to the Dynamic mask-based modulation, which enables the dynamic pruning of redundant sampling points. Based on these results, we typically find a radius of 3 or 5 to be optimal.

4.3.3 Differentiable shuffles. As shown in Table 4, we compare our method against differentiable ranking approaches that approximate rank using probability distributions over element order.

Our method avoids gradient bias by directly learning the order via global indicators without transforming the optimization target, whereas other methods, such as soft ranking, introduce this bias by replacing the original target with an approximate probabilistic one.

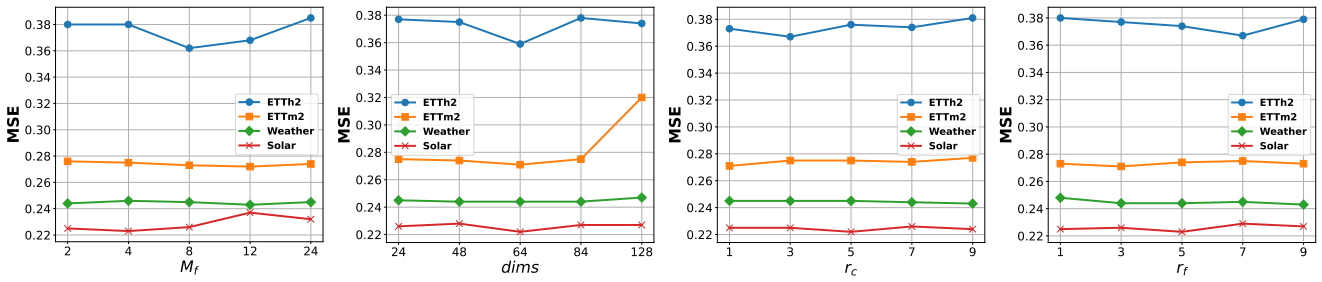
By avoiding the introduction of noise and uncertainty and by generating a more robust ranking from global indicators, our *Adaptive 2-Dimensional Shuffle* achieves superior performance. This advantage is particularly pronounced on the small-sample ETTh2 dataset, which is more sensitive to such noise.

Table 4: Comparison of differentiable shuffles.

Datasets	ETTh2		ETTm2		Weather		Solar	
Metrics	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
without gradient	0.386	0.404	0.285	0.329	0.258	0.282	0.238	0.239
soft rank [69]	0.368	0.391	0.273	0.315	0.247	0.267	0.226	0.226
fast rank [2]	0.372	0.393	0.275	0.316	0.245	0.266	0.228	0.227
ours	**0.362**	**0.389**	**0.270**	**0.313**	**0.241**	**0.263**	**0.218**	**0.222**

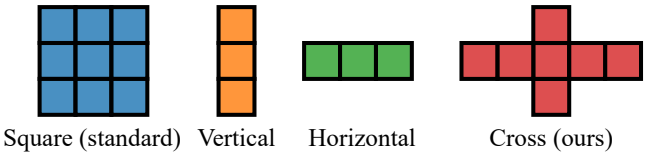
4.3.4 Initialization shape. To further demonstrate that our *Cross-Extension initialization* is effective for *Channel-Scale* and *Channel-Frequency* view with image-like structures, we explored the impact of different initialization shapes (a. Square shape, b. Vertical line, c. Horizontal line, d. Cross shape, as shown in Figure 5a) on the deformable convolution. For a fair comparison, we remove the *Channel-Scale* view branch (as the two views have dissimilar sizes) and implement the uniform sampling strategy. Figure 5b illustrates the impact of the four initialization shapes on the model's predictive performance as the number of sampling points $num \in (0, 40)$. We select Solar, which comprises 137 channels, for our experiments and set both the input length T and the forecast horizon F to 96.

We have the following observations: 1) Square initialization provides a broad local receptive field, making it highly effective when the number of sampling points is limited (e.g., $num = 4$).

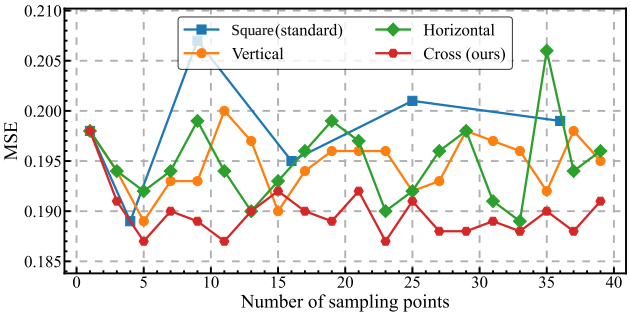


(a) Frequency patch numbers (b) Hidden dimension size (c) Channel radius (d) Frequency radius

Figure 4: Parameter sensitivity studies of main hyper-parameters in CCD.



(a) Initialization shapes



(b) MSE of different sampling numbers

Figure 5: Comparison of different initialization shapes.

However, the Channel-Frequency view presents two fundamental challenges: it lacks geometric properties and is inherently asymmetric. Consequently, as the number of points increases, this method incorporates more noise, leading to a significant performance drop (e.g., to $num = 9$). 2) Vertical and Horizontal initializations are more inclined to capture *Intra-Correlations*. As the number of sampling points increases, their sampling range expands, but only in a single direction. This unidirectional extension results in the loss of considerable valuable information. 3) Our method, using a cross-shaped initialization, can independently and asymmetrically expand the receptive field along two dimensions. The deformable sampling points are initialized at the locations of *Intra-Correlations*, and their offsets are then learned to capture key *Cross-Correlations*. The results show that our method achieves optimal performance and demonstrates strong robustness.

4.3.5 Comparison against Advanced Deformable Convolutions. To further validate the effectiveness of our approach, we compare it against standard deformable convolution DCN [17] and advanced variants, including DSConv [52], FDNet [40], and DCNv3 [76]. The results are summarized in Table 5.

Table 5: Comparison against deformable convolutions.

Dataset	ETTm2		Weather		Solar	
Metric	MSE	MAE	MSE	MAE	MSE	MAE
DCN	0.283	0.324	0.285	0.297	0.268	0.265
DCNV3	0.284	0.329	0.281	0.299	0.263	0.260
DSConv	0.281	0.319	0.287	0.293	0.255	0.257
FDNet	0.275	0.320	0.283	0.294	0.257	0.259
Cross DConv	**0.270**	**0.313**	**0.241**	**0.263**	**0.218**	**0.222**

As observed, directly applying existing deformable convolution variants to time series views yields limited and unstable improvements. Although methods such as DCNv3 and DSConv are highly flexible, they are originally designed for image-like spatial structures, where neighboring positions usually have clear geometric meanings. In contrast, our constructed time series views are organized along channel-frequency or channel-scale dimensions, whose local neighborhoods do not naturally follow the same spatial assumptions as images. Therefore, their deformable sampling may easily attend to less informative or weakly related positions.

5 Conclusion

In this study, we propose CCD, a novel framework for multivariate time series forecasting that captures complex *Cross-Correlations* across variables, frequencies, and scales. Specifically, CCD transforms time series into *Channel-Frequency* and *Channel-Scale* views, and introduces an *Adaptive 2-Dimensional Shuffle* module to rearrange them into more structured representations. To further model cross-view dependencies, we design *Cross Deformable Convolution*, enhanced by *Non-uniform Cross-Extension initialization* and *Dynamic mask-based modulation*. By integrating these components, CCD effectively captures cross-dimensional correlations and achieves state-of-the-art forecasting performance.

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A Correlation Metric Details

To visualize the distribution and intensity of *Cross-Correlations*, we have developed separate methods for measuring inter-channel dependencies in the Channel-Frequency and Channel-Scale views, accounting for their inherent differences.

A.1 *Cross-Correlations* in *Channel-Scale* view

The Pearson correlation coefficient is a common means of measuring inter-channel correlation, but it cannot be used directly due to the inconsistent sequence lengths between different scales in the *Channel-Scale* view. The *Cross-Correlations* coefficient can handle sequences of different lengths through a sliding dot product operation, but its main function is to measure shape matching rather than a statistical linear relationship.

We combine the advantages of both to construct the Maximum Lagged Correlation, which can measure the linear correlation between sequences of different lengths. The specific method is shown in Algorithm 1.

Algorithm 1 Maximum Lagged Correlation

Input: The series of channel i at scale m X_i^m and another series of channel j at scale n X_j^n.
Output: The maximum lagged Pearson correlation value $c_{\max}$.

If $|X_i^m| > |X_j^n|$ **then**
  $s_{\text{short}} \leftarrow X_j^n,\ s_{\text{long}} \leftarrow X_i^m$
Else
  $s_{\text{short}} \leftarrow X_i^m,\ s_{\text{long}} \leftarrow X_j^n$
EndIf
$L_{\text{short}} \leftarrow |s_{\text{short}}|,\ L_{\text{long}} \leftarrow |s_{\text{long}}|$
$C \leftarrow []$
For $k = 0$ to $L_{\text{long}} - L_{\text{short}}$ **do**
  $w \leftarrow s_{\text{long}}[k : k + L_{\text{short}}]$
  $c \leftarrow \text{PearsonCorr}(s_{\text{short}}, w)$
  Append c to C
EndFor
If C is empty **then return** 0.0
$idx \leftarrow \underset{l}{\text{argmax}}\,|C[l]|$
$c_{\max} \leftarrow C[idx]$
return $c_{\max}$

A.2 *Cross-Correlations* in *Channel-Frequency* view

In order to better measure the correlation between sequences in the frequency domain, we use a complex generalization of the Pearson correlation coefficient. This method not only measures the strength of the association between two complex sequences but also reflects their average phase difference. The specific method is shown in Algorithm 2.

Algorithm 2 Complex Correlation Coefficient

Input: The spectrum of channel i at m-th frequency patch X_i^m the spectrum of channel j at n-th frequency patch X_j^n.
Output: The complex correlation coefficient $c \in \mathbb{C}$.

$\mu_{i,m} \leftarrow \text{Mean}(X_i^m), \quad \mu_{j,n} \leftarrow \text{Mean}(X_j^n)$
$a \leftarrow X_i^m - \mu_{i,m}$
$b \leftarrow X_j^n - \mu_{j,n}$
$N \leftarrow \sum (a \cdot \text{conj}(b))$
$D \leftarrow \sqrt{\sum |a|^2} \cdot \sqrt{\sum |b|^2}$
If $D \approx 0$ **then return** $0 + 0i$
return N / D

B Limitations

Although CCD effectively captures *Cross-Correlations* and achieves state-of-the-art performance via the Adaptive 2D Shuffle and Cross DConv modules, we acknowledge the following limitations in practical applications:

- **Edge Deployment:** While CCD exhibits high efficiency and a moderate parameter count on GPUs, the irregular memory access patterns inherent to Deformable Convolution may pose optimization challenges and impose additional computational overhead on resource-constrained edge devices.
- **Interpretability:** Although the *Adaptive 2-Dimensional Shuffle* significantly enhances the receptive field efficiency of CNNs, the reordered latent feature maps disrupt the original topological structure of the data. This makes it challenging to directly derive intuitive interpretations from the intermediate layers.
- **Extremely Short Sequences:** CCD relies on channel-frequency and channel-scale views generation. When input sequences are extremely short, effective scale and frequency decomposition becomes difficult, which diminishes the richness of information available for Cross-Correlation modeling.